

THE PLACE OF FAST REACTORS IN THE FRENCH NUCLEAR PROGRAMME

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FAST reactors occupy an important and growing place in the French nuclear programme. As an indication, the number of people working on them full time in the Commissariat à l'Energie Atomique at the Cadarache, Saclay and Fontenay-aux-Roses centres is of the order of 500, one-third of whom are engineers, which represents about 20% of the number of staff working on power producing reactors. Electricité de France also demonstrate the very great interest which they have in this reactor system by backing up the action of the CEA with a contribution in men and test facilities.

Fast reactors permit the fullest use to be made of the fissile material resources, but it is only in the long term that this possibility will become an economic necessity. We consider that the potential advantages of breeders should be capable of ensuring, in any case, that they are economically viable, even before any substantial re-enrichment of the uranium gives them additional value.

Breeding constitutes *a priori* a very favourable element in an economic fuel cycle; it is also necessary to develop a fuel permitting long irradiation (about 100,000 MWd/t) and achieve cheap reprocessing.

The great compactness and the possibility of obtaining a high efficiency are also features which favour a reduction in capital costs.

Fast reactors will benefit more from these features, the higher is their power. It can now be foreseen that in the future they will be used in base-load stations with a unit capacity of at least 1000 MW(e).

It may be recalled that between now and the end of the fifth 5-year plan (1966-70) France will have started the installation of nuclear capacity of between 4500 and 6000 MW(e). The majority will be in the form of natural

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uranium/graphite reactors which are excellent converters (producing about 400 g of fissile plutonium per year per MW(e) installed) and thus prepare the way quite naturally to fast breeders.

Around 1975 the annual production of plutonium in France will be between 1 and 2 tonnes. The construction of the first fast reactor station of 1000 MW(e) can therefore be envisaged between 1975 and 1980 (reckoning on about 3.5 kg of fissile plutonium per MW(e) installed, 2.5 kg of which is in the reactor core) and this prospect gives us the time to resolve the technical problems which exist at the moment.

For this reactor system, as for the others, the problem is to find the happy medium between excessive caution, which would hinder its development, and premature attempts which could cause delays or excessive costs.

In view of the existence of Rapsodie and all the studies and tests which accompanied it, we think we have acquired the technological experience essential for taking the next step.

At the point we have reached we consider that the safest way to arrive at the creation of a 1000 MW(e) fast reactor station between 1975 and 1980 is first to build a prototype of a significant capacity.

Studies on this prototype, called Phenix, started at the beginning of 1965.

Basically we expect Phenix to serve as a technical and industrial demonstration, i.e. provide proof that we can build and operate a commercial fast reactor without encountering any major obstacle in the technical sphere. This idea will lead us to adopt solutions which are as well tried, safe and simple as possible. On the other hand it will cause us not to expect exceptional performance, considering that there will always be time on the later reactors, or perhaps even on subsequent cores in Phenix, to make improvements when the experience acquired justifies it.

Our second preoccupation is for Phenix to be representative of the system, since it is only a stage on the way to the large breeder stations, which one can expect to be developed industrially on a large scale after 1980. From the very beginning we were anxious for it to pave the way for the future and well before starting the studies on Phenix itself we began, 3 years ago, by exploring the characteristics of reactors of 1000 MW(e) so that they would serve as some sort of guide. Therefore we shall ensure that the technical solutions used for Phenix can be extrapolated to the scale of future stations, to the extent that they are compatible with the idea of reliability expressed above.

For our studies we had laid down initially two reference capacities (125 or 250 MW(e)) corresponding to the stages adopted by Electricité de France for its thermal stations. We soon placed the emphasis on the 250 MW(e) version which provides a much better foundation for a 1000 MW(e) station and permits the use of the existing turbo-alternator sets by producing superheated steam at 540°C at a pressure of 130×10^4 kg/cm².

The Phenix studies are being undertaken with a view to obtaining the

maximum benefit from the experience gained with Rapsodie, the construction of which is being completed at Cadarache and which we hope to operate at its rated power during 1967.

The power of Rapsodie is only 20 MW(th) but in many respects its characteristics are comparable with those of commercial reactors. Thus we are aiming at a specific power of 500 kW/litre in the centre and a sodium outlet temperature between 500°C and 550°C. The mechanical and engineering components have been designed so that as far as possible they can be extrapolated to reactors of a much higher power. To take just one example we can quote the choice of mechanical centrifugal pumps on the primary and secondary circuits in preference to electromagnetic pumps.

Phenix, like Rapsodie, will use sodium as the heat transfer fluid. Without disputing the possibility of building fast reactors using other fluids (gas or steam) we continue to believe that sodium is the most suitable for their requirements. The technological experience which we have acquired with it, during more than 10 years' research, in particular on the test facilities for Rapsodie, gives us every confidence in the future of sodium reactors. To avoid spreading our effort we decided to place all our means at the disposal of this basic system.

The only sphere of sodium technology in which Rapsodie provides us with nothing is the very important one of the steam generators. To deal with the problems we preferred to build specially for this purpose an experimental loop of 5 MW(th) installed at Grand Quevilly, which has been producing steam at 540°C and 130×10^4 kg/cm² since the end of 1964. It is on this loop that we shall develop the type to be used for Phenix after testing several systems.

In addition, Electricité de France have just decided to design a steam generator test facility of a higher capacity, in order to permit the development of this technique.

The same desire for continuity exists in regard to the fuel. At the beginning of 1963 we chose mixed oxide UO₂-PuO₂ in 316 stainless steel sealed cans for Rapsodie, which at the time seemed the most promising system. This judgement has not been contradicted by the results of tests carried out since then in various countries. For the first core of Phenix we are using the same type of fuel as the reference fuel. By doing this we shall benefit fully from the previous fabrication experience and even from the operating results of Rapsodie which will serve by priority as an irradiation test bed.

An extensive research and test programme has been launched to enable us to lay down in the middle of 1970 the final specifications for the fuel for the first core in Phenix. At the same time more long-term research is being carried out. This covers basically:

- (1) studies on fuel elements using carbides or mixtures of carbides and nitrides of uranium and plutonium;

- (2) studies on canning materials other than stainless steel: nickel and vanadium alloys.

The development of satisfactory vented cans would constitute a great advance in the development of fast reactor fuel technology. It is our intention to test these first in Rapsodie.

We shall not go into detail here on the preparatory studies for defining the characteristics of Phenix. Other papers at this conference will describe these, particularly in regard to the overall design of the reactor, the mechanical layout of the reactor block, the primary circuits and handling equipment and the work on the main components.

TABLE 1. MAIN CHARACTERISTICS OF THE PHENIX CORE
(Provisional values)

Core volume (litres)	1060
Core height (cm)	70
Proportion of core volume occupied by fuel (%)	42
Fuel	depleted UO ₂ + PuO ₂
Pellet diameter (mm)	5
Critical mass (kg of fissile plutonium)	615
Canning material	AISI 316 stainless steel
Can thickness (mm)	0.40
Can hot spot temperature (°C)	700
Number of pins per assembly	271
Total number of fuel assemblies	120
Pressure drop	7×10^4 kg/cm ²
Power (MW(e))	250
Method of power flattening	two zones with different enrichment
Mean specific power:	
kW/litre of core	565
MW(th)/kg fissile Pu	0.975
Maximum linear power, nominal at half-life (W/cm)	380
Temperature difference between primary sodium outlet and inlet (°C)	160
Mean primary sodium outlet temperature (°C)	560
Steam temperature (°C)	540

We shall simply give a table (Table 1) summarizing the main characteristics of the first core of Phenix in its 250 MW(e) version. At the present stage of our research, it has only been provisionally designed. Even if they are only an indication, the values contained in this table nevertheless give a good idea of what the final core will be.

The whole of our research and test programme is being carried out with the firm intention of commencing the construction of Phenix at the beginning of 1969.

We should recall that the first stage in the development of fast reactors in France (comprising Rapsodie, Harmonie, Masurca and the studies on the system) is being carried out within a Contract of Association between Euratom and the CEA concluded in 1962 for 5 years.